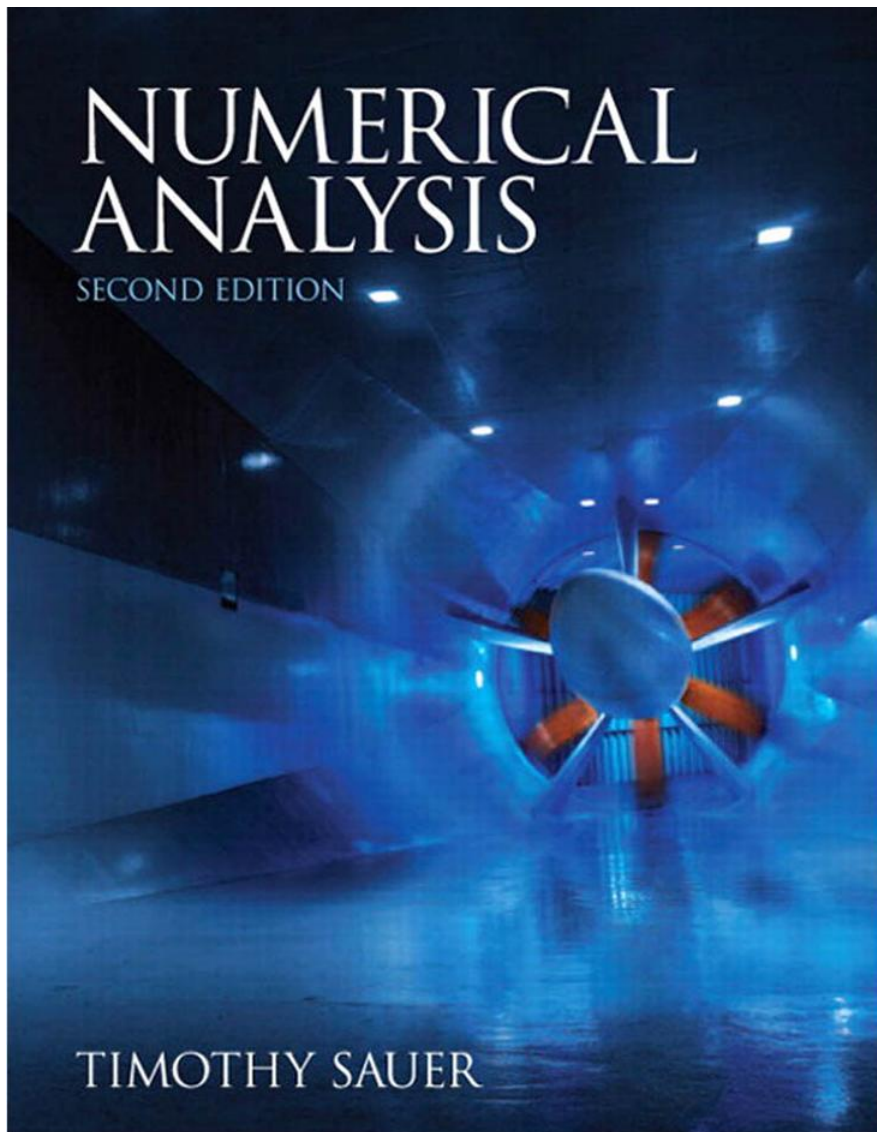


NUMERICAL ANALYSIS

SECOND EDITION

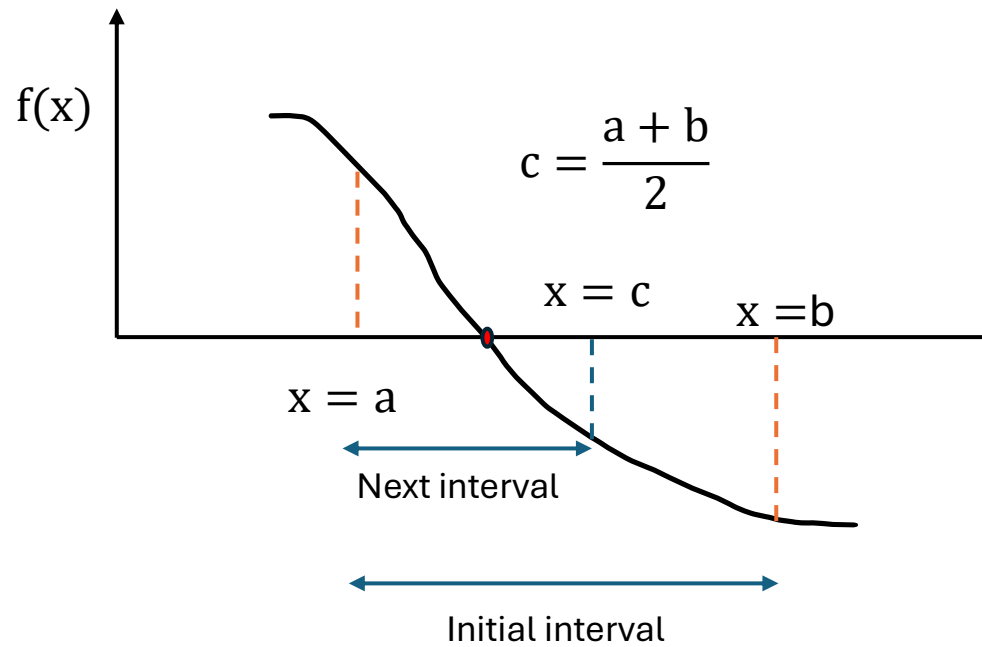
TIMOTHY SAUER



Root Finding Methods

Bisection Method:

$$f(x) = 0$$

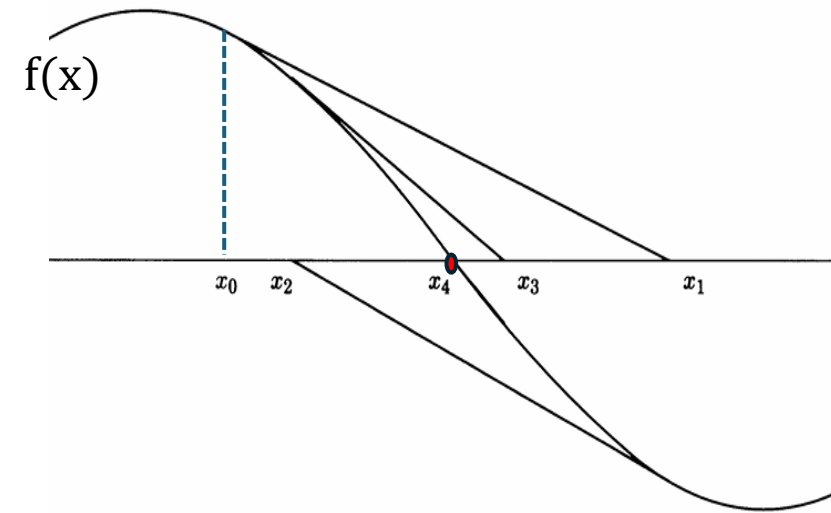


Newton-Raphson Method:

$$f'(x_0)(x_1 - x_0) = 0 - f(x_0)$$

$$x_1 - x_0 = -\frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$



Linear convergence of Newton's Method

Use Newton's Method to find a root of $f(x) = x^2$.

$$\begin{aligned}x_{i+1} &= x_i - \frac{f(x_i)}{f'(x_i)} \\&= x_i - \frac{x_i^2}{2x_i} \\&= \frac{x_i}{2}.\end{aligned}$$

$x_i - r = e_i =$ error at the i th iteration

$r = 0 =$ root of the polynomial x^2

$$e_{i+1} = e_i/2$$

Similarly, for $f(x) = x^m$

$$\begin{aligned}x_{i+1} &= x_i - \frac{x_i^m}{mx_i^{m-1}} \\&= \frac{m-1}{m}x_i.\end{aligned}$$

$$e_{i+1} = Se_i,$$

where $S = \frac{m-1}{m}$

Multiplicity of roots:

$f(x) = (x - r)^m Q(x)$ The root r has a multiplicity of m

$$f(x) = x^2 - 2x + 1 = (x - 1)^2 \text{ (root 1 has a multiplicity of 2)}$$

Find the multiplicity of the root $r = 0$ of $f(x) = \sin x + x^2 \cos x - x^2 - x$

$$f(x) = \sin x + x^2 \cos x - x^2 - x$$

$$f'(x) = \cos x + 2x \cos x - x^2 \sin x - 2x - 1$$

$$f''(x) = -\sin x + 2\cos x - 4x \sin x - x^2 \cos x - 2$$

$$f'''(x) = -\cos x - 6\sin x - 6x \cos x + x^2 \sin x,$$

$$f(0) = 0, f'(0) = 0, f''(0) = 0, f'''(0) = -1$$

Modified Newton-Raphson Method:

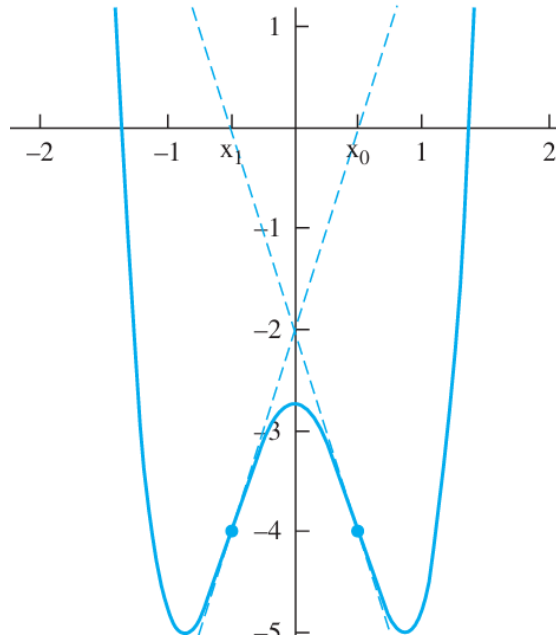
$$x_{i+1} = x_i - \frac{m f(x_i)}{f'(x_i)}$$

This has a quadratic convergence to the root, i.e.,

$$e_{i+1} \propto e_i^2 \text{ (Exercise)}$$

Failure of convergence of the Newton-Raphson Method:

Apply Newton's Method to $f(x) = 4x^4 - 6x^2 - 11/4$ with starting guess $x_0 = 1/2$.



Initial-condition dependent

$$\begin{aligned} f'(x_0)(x_1 - x_0) &= 0 - f(x_0) \\ x_1 - x_0 &= -\frac{f(x_0)}{f'(x_0)} \\ x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)}. \end{aligned}$$

The method fails if $f'(x)=0$

Secant Method:

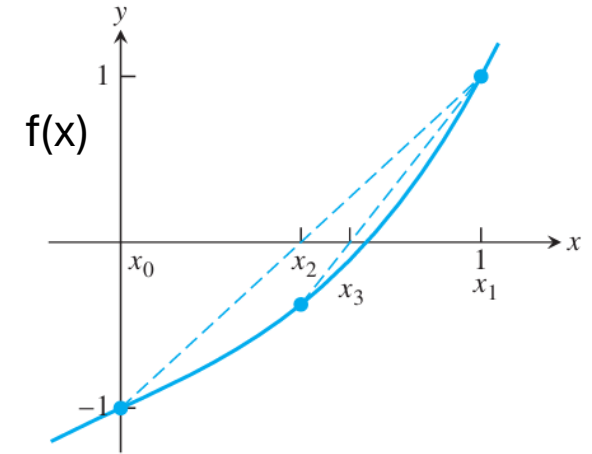
$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

(Newton's Method)



$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

$x_0, x_1 = \text{initial guesses}$

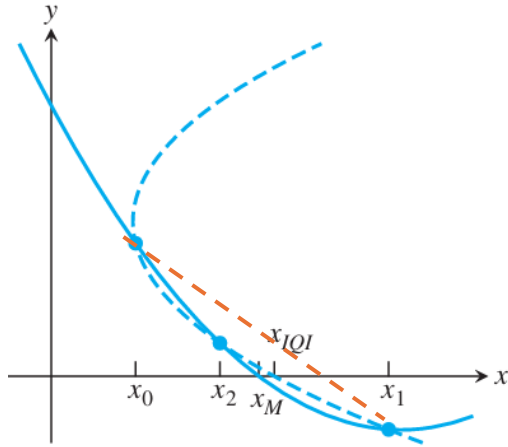


Convergence:

$$e_{i+1} \approx \left| \frac{f''(r)}{2f'(r)} \right| e_i e_{i-1} \quad \Rightarrow \quad e_{i+1} \approx \left| \frac{f''(r)}{2f'(r)} \right|^{\alpha-1} e_i^\alpha,$$

where $\alpha = (1 + \sqrt{5})/2 \approx 1.62$.

Inverse Quadratic Interpolation (IQI)



$$y = f(x) \quad x = P(y)$$

The second-degree polynomial $x = P(y)$ that passes through the three points (a, A) , (b, B) , (c, C) is

(Lagrange Interpolation)

$$P(y) = a \frac{(y - B)(y - C)}{(A - B)(A - C)} + b \frac{(y - A)(y - C)}{(B - A)(B - C)} + c \frac{(y - A)(y - B)}{(C - A)(C - B)}.$$

$$P(A) = a, P(B) = b, \text{ and } P(C) = c.$$

$$P(0) = c - \frac{r(r - q)(c - b) + (1 - r)s(c - a)}{(q - 1)(r - 1)(s - 1)},$$

For IQI, after setting $a = x_i, b = x_{i+1}, c = x_{i+2}$, and $A = f(x_i), B = f(x_{i+1}), C = f(x_{i+2})$, the next guess $x_{i+3} = P(0)$ is

where $q = f(x_i)/f(x_{i+1}), r = f(x_{i+2})/f(x_{i+1})$, and $s = f(x_{i+2})/f(x_i)$

$$x_{i+3} = x_{i+2} - \frac{r(r - q)(x_{i+2} - x_{i+1}) + (1 - r)s(x_{i+2} - x_i)}{(q - 1)(r - 1)(s - 1)},$$

Convergence of Secant Method:

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

$$x_i = r + e_i$$

$$e_{i+1} = e_i - \frac{f(r + e_i)(e_i - e_{i-1})}{f(r + e_i) - f(r + e_{i-1})}$$

$$f(r + e_i) = f(r) + e_i f'(r) + \frac{e_i^2}{2} f''(r)$$

$$f(r + e_i) = e_i f'(r) + \frac{e_i^2}{2} f''(r) \quad [f(r)=0]$$

$$\begin{aligned} f(r + e_i) - f(r + e_{i-1}) &= (e_i - e_{i-1})f'(r) + \frac{e_i^2 - e_{i-1}^2}{2} f''(r) \\ &= (e_i - e_{i-1})f'(r) \left[1 + \frac{(e_i + e_{i-1})}{2f'(r)} f''(r) \right] \\ &= (e_i - e_{i-1})f'(r) [1 + (e_i + e_{i-1})C] \end{aligned}$$

$$\text{where } C = \frac{f''(r)}{2f'(r)}$$

$$e_{i+1} \approx e_i - \frac{e_i f'(r)(1 + Ce_i)(e_i - e_{i-1})}{(e_i - e_{i-1})f'(r)[1 + (e_i + e_{i-1})C]} \approx Ce_i e_{i-1}$$

$$|e_{i+1}| \approx K|e_i^\alpha| \quad (1)$$

$$K|e_i^\alpha| \approx C|e_i| |e_{i-1}|$$

$$K|e_i|^{\alpha-1} \approx C|e_{i-1}|$$

$$|e_i| \approx \left[\frac{C}{K} \right]^{1/(\alpha-1)} |e_{i-1}|^{1/(\alpha-1)} \quad (2)$$

Comparing (1) and (2)

$$\begin{aligned} \frac{1}{(\alpha-1)} = \alpha &\Rightarrow \alpha^2 - \alpha - 1 = 0 \\ &\Rightarrow \alpha = \frac{1 + \sqrt{5}}{2} \end{aligned}$$

$$|e_{i+1}| \approx C^{\alpha-1} |e_i|^\alpha$$